ELE-plan for i dag

← *Lidt rep. med negativ feedback?*

Op Amp fejl i det lineære område

- *Frekvensafhængighed i A_{OL} og A_{CL}*
- *Hvorfor er frekvenskompensering nødvendig?*
- *Hvordan opnås frekvenskompensering i praksis?*

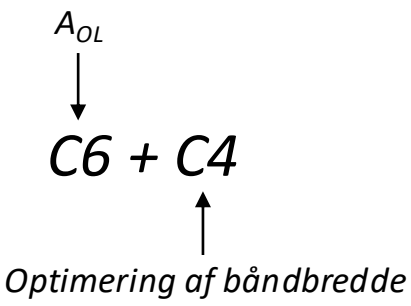
Mætningsfænomener

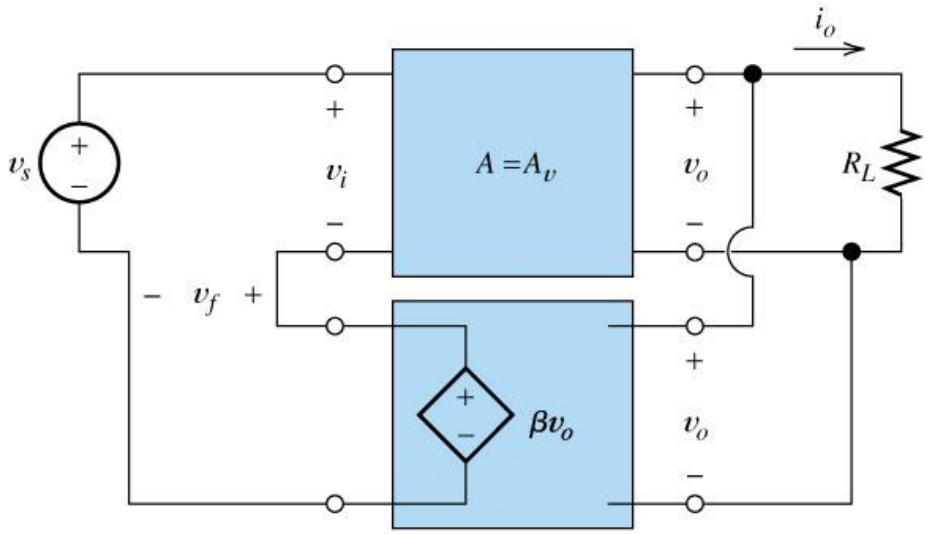
- *En begrænsning ud over den dynamiske båndbredde*

LF356/357

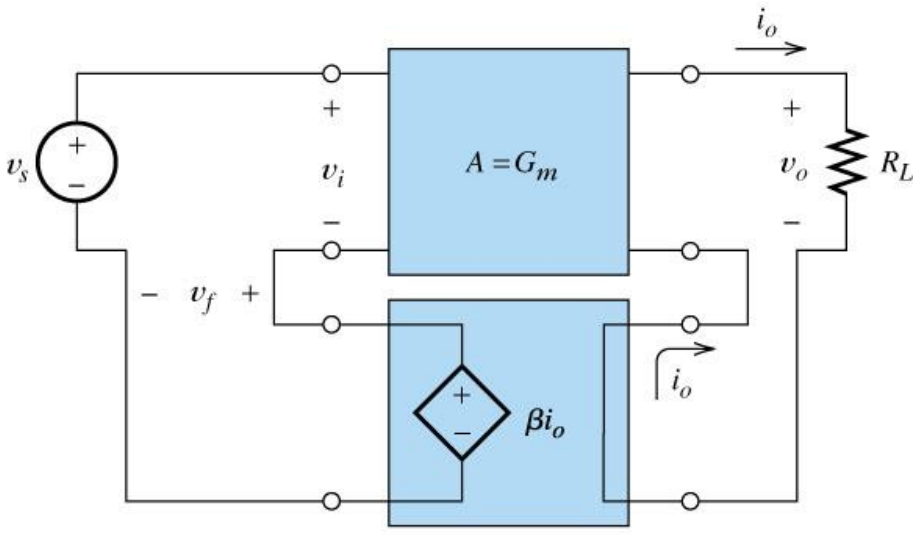
- *Et kig indenfor i en Op Amp* ☺

Eksempler og opgaver

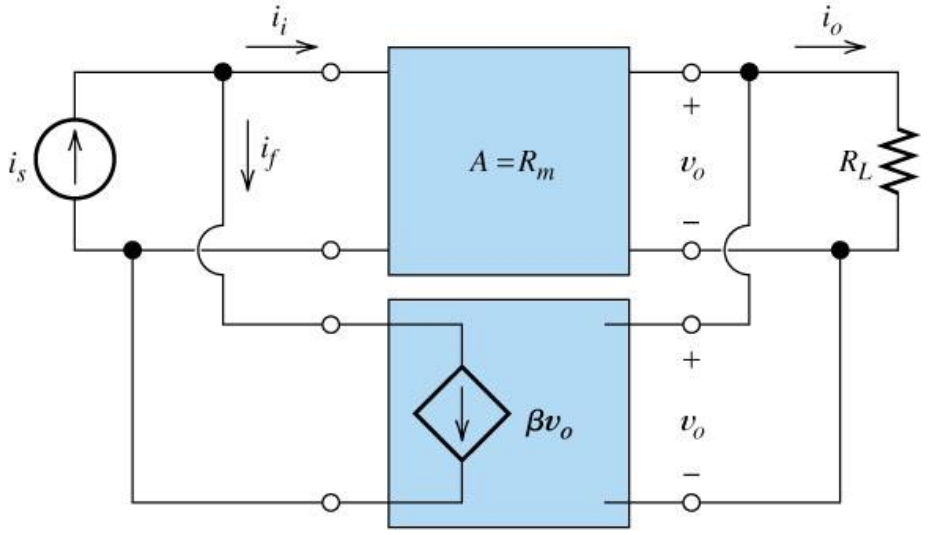




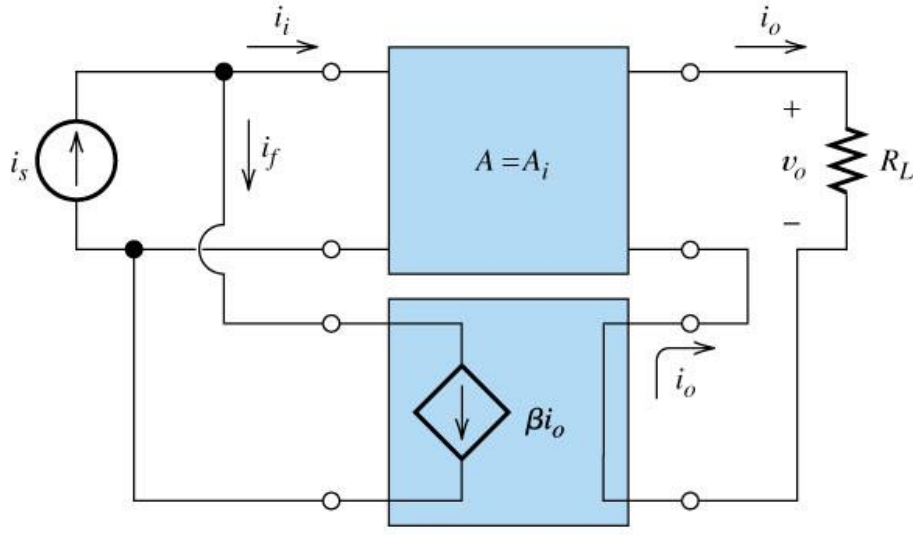
(a) Series voltage feedback



(b) Series current feedback



(c) Parallel voltage feedback



(d) Parallel current feedback

Figur 9.14

Table 9.1. Effects of Feedback[a]

Feedback Type	x_s	x_o	Gain Stabilized	Input Impedance	Output Impedance	Ideal Amplifier
Series voltage	v_s	v_o	$A_{vf} = \dfrac{A_v}{1 + A_v\beta}$	$R_i(1 + A_v\beta)$	$\dfrac{R_o}{1 + \beta A_{voc}}$	Voltage
Series current	v_s	i_o	$G_{mf} = \dfrac{G_m}{1 + G_m\beta}$	$R_i(1 + G_m\beta)$	$R_o(1 + \beta G_{msc})$	Transconductance
Parallel voltage	i_s	v_o	$R_{mf} = \dfrac{R_m}{1 + R_m\beta}$	$\dfrac{R_i}{1 + R_m\beta}$	$\dfrac{R_o}{1 + \beta R_{moc}}$	Transresistance
Parallel current	i_s	i_o	$A_{if} = \dfrac{A_i}{1 + A_i\beta}$	$\dfrac{R_i}{1 + A_i\beta}$	$R_o(1 + \beta A_{isc})$	Current

[a] Formulas given assume an ideal controlled source for the feedback network (as shown in Figure 9.14), zero source impedance for series feedback, and infinite source impedance for parallel feedback. Gains with subscripts sc and oc are for short-circuit and open-circuit loads, respectively. The gains A_v, G_m, R_m, and A_i are for the actual load.

Table 9.1. Effects of Feedback[a]

Feedback Type	x_s	x_o	Gain Stabilized	Input Impedance	Output Impedance	Ideal Amplifier
Series voltage	v_s	v_o	$A_{vf} = \dfrac{A_v}{1 + A_v\beta}$	$R_i(1 + A_v\beta)$	$\dfrac{R_o}{1 + \beta A_{voc}}$	Voltage
Series current	v_s	i_o	$G_{mf} = \dfrac{G_m}{1 + G_m\beta}$	$R_i(1 + G_m\beta)$	$R_o(1 + \beta G_{msc})$	Transconductance
Parallel voltage	i_s	v_o	$R_{mf} = \dfrac{R_m}{1 + R_m\beta}$	$\dfrac{R_i}{1 + R_m\beta}$	$\dfrac{R_o}{1 + \beta R_{moc}}$	Transresistance
Parallel current	i_s	i_o	$A_{if} = \dfrac{A_i}{1 + A_i\beta}$	$\dfrac{R_i}{1 + A_i\beta}$	$R_o(1 + \beta A_{isc})$	Current

[a] Formulas given assume an ideal controlled source for the feedback network (as shown in Figure 9.14), zero source impedance for series feedback, and infinite source impedance for parallel feedback. Gains with subscripts sc and oc are for short-circuit and open-circuit loads, respectively. The gains A_v, G_m, R_m, and A_i are for the actual load.

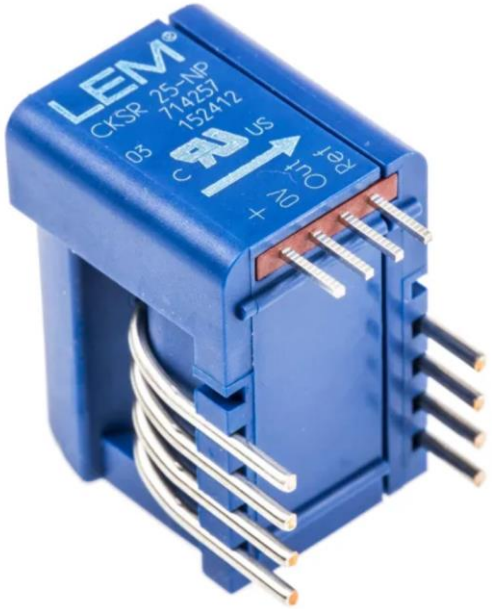
$A_v \sim G_m \sim R_m \sim A_i \sim A_{OL}$

Strømtransformer

LEM Strømtransformer -40°C -> +105°C 25A 25:1

RS-varenummer: **666-8205** | Producentens varenummer: **CKSR 25-NP** | Brand: LEM



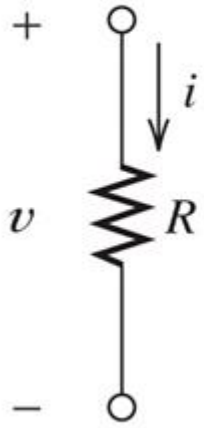


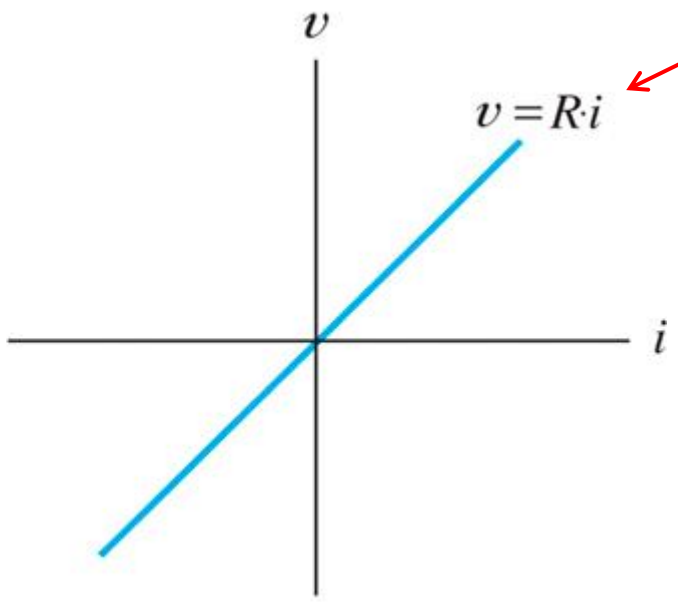
[Link til RS]

Modstand - måske et bedre alternativ?



*Den ideelle modstand er kendetegnet ved at **spændingen over** den er proportional med **strømmen gennem** den.*

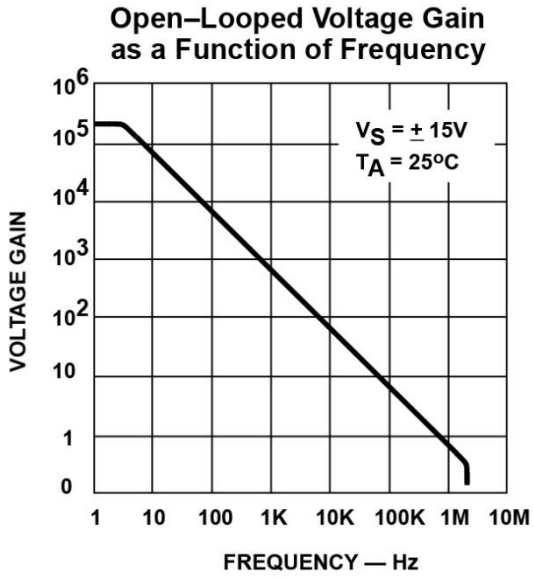




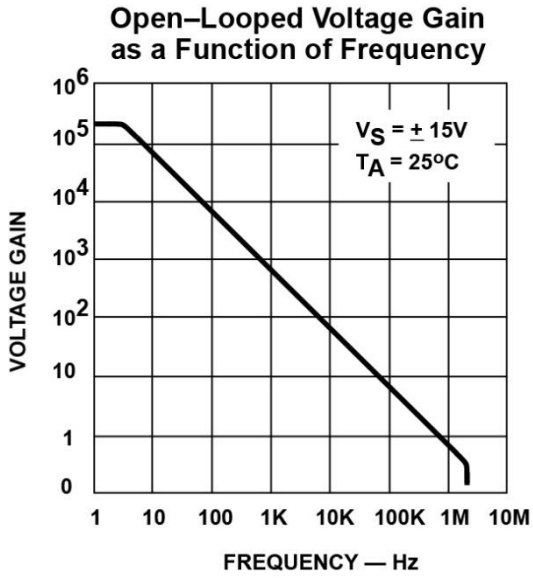
Ohms lov

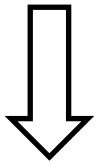

A_{OL}

A_{OL} *frekvensafhængig*



A_{OL} *frekvensafhængig*

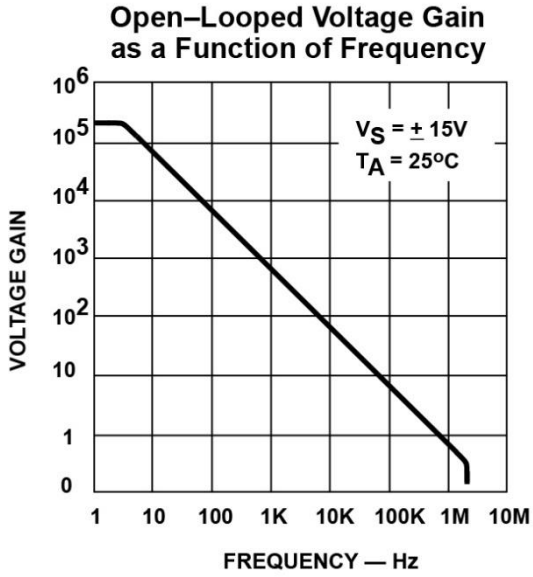




K_f *frekvensafhængig:*

$$K_f = \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}}$$

A_{OL} *frekvensafhængig*



⇩

K_f *frekvensafhængig:*

$$K_f = \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}}$$

⇩

A_{CL} *frekvensafhængig:*

$$A_{CL} = \frac{\alpha}{\beta} \cdot K_f$$

Amplitude og fase



Kompleks overføringsfunktion…

Åbensløjfe forstærkningen A_{OL}

$$A_{OL}(f) = \frac{A_{0OL}}{1+j\frac{f}{f_{BOL}}} \qquad (2.24)$$

A_{0OL} : Åbensløjfe DC forstærkning, dvs. $A_{0OL} = A_{OL}(0)$

f_{BOL} : Åbensløjfe knækfrekvens

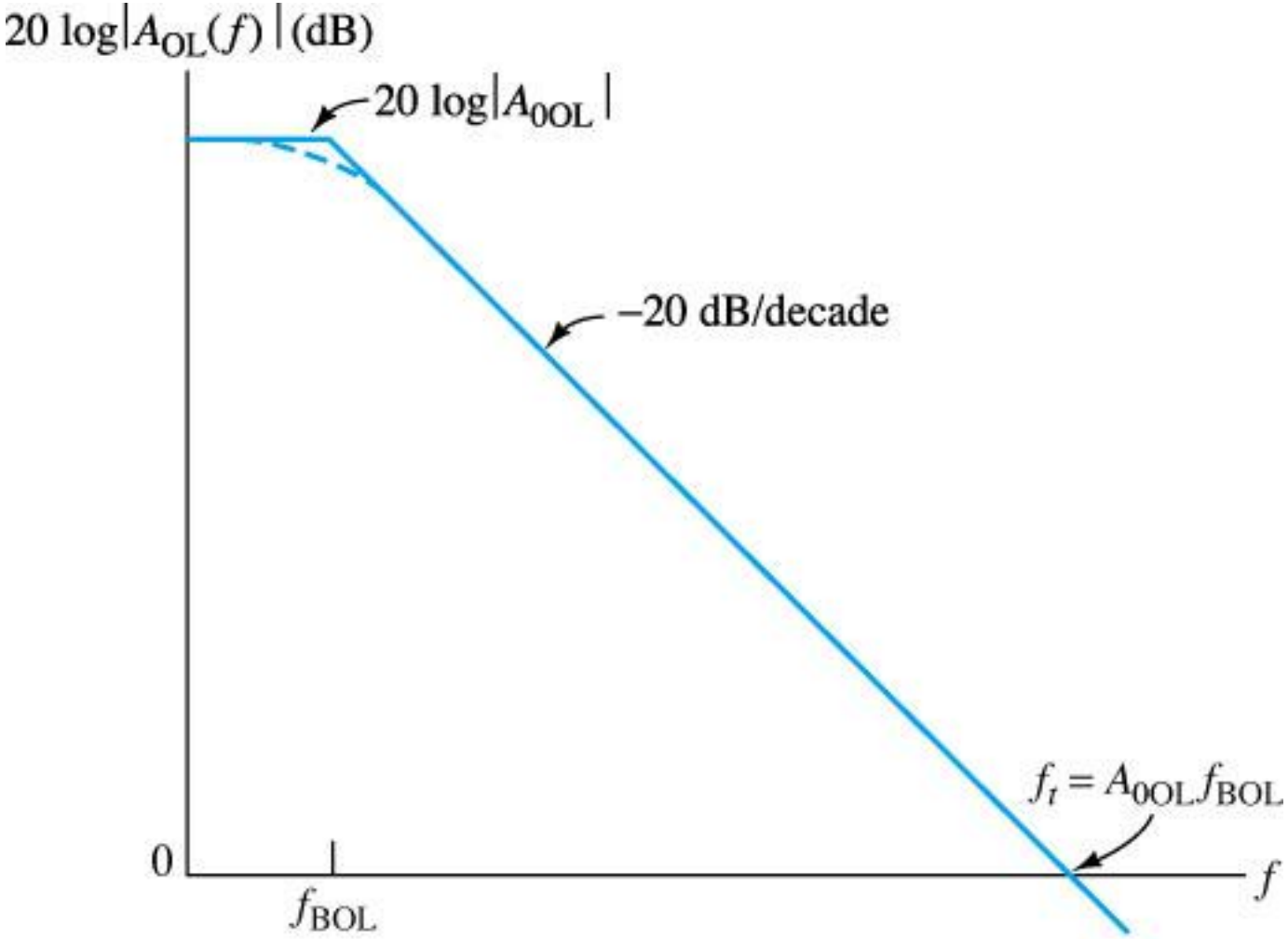


Figure 2.25 Bode plot of open-loop gain for a typical op amp.

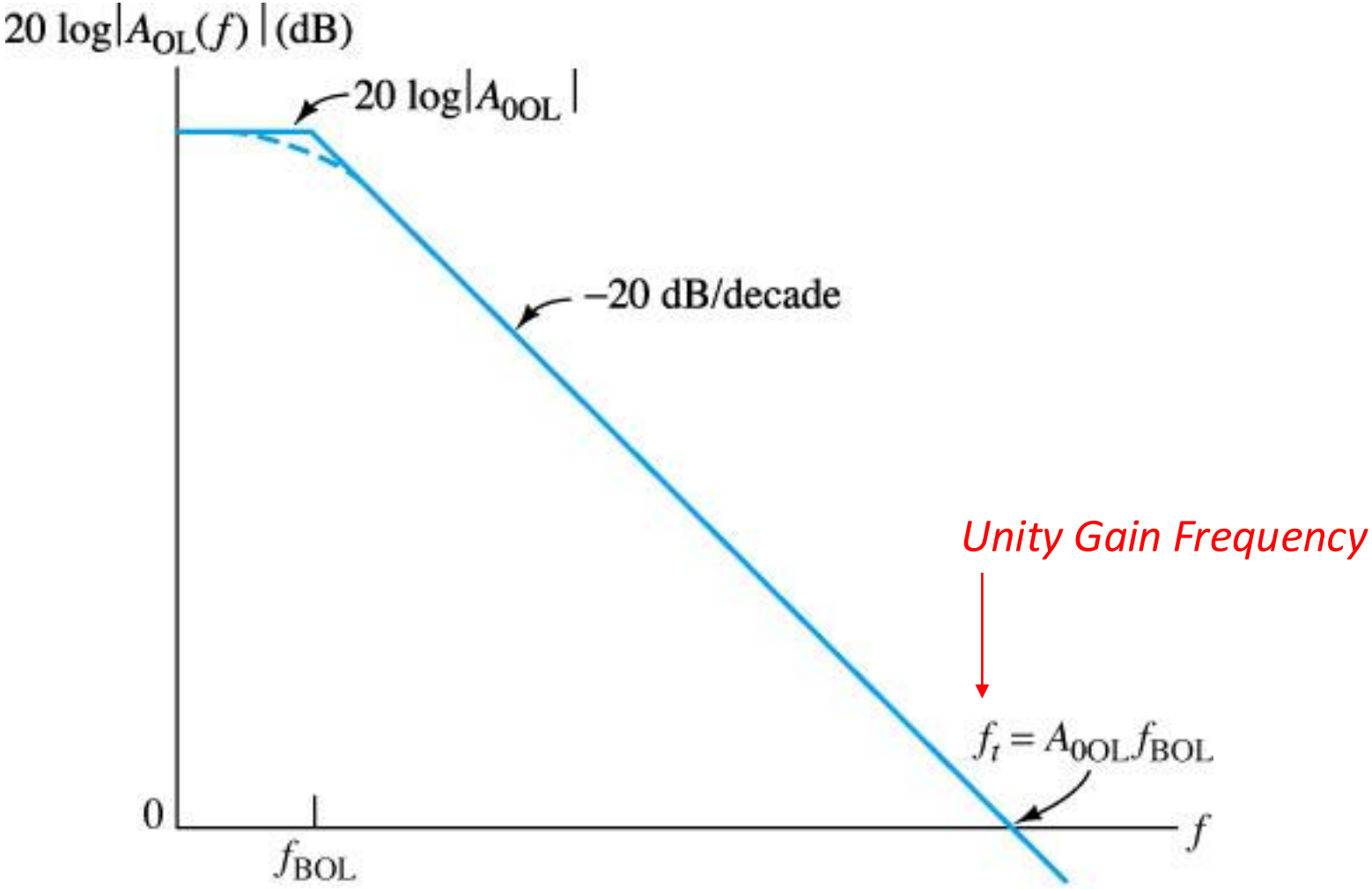


Figure 2.25 Bode plot of open-loop gain for a typical op amp.

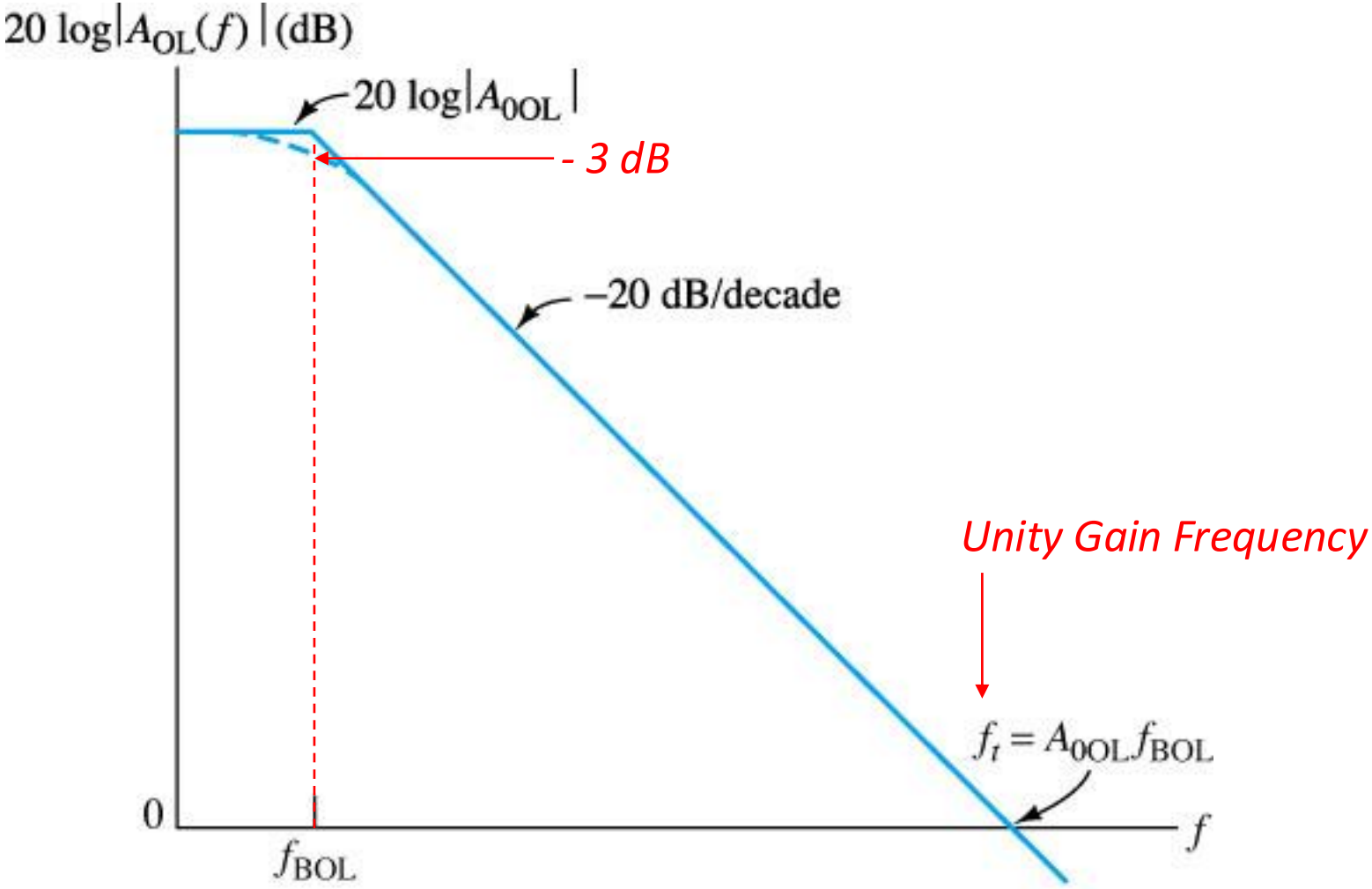
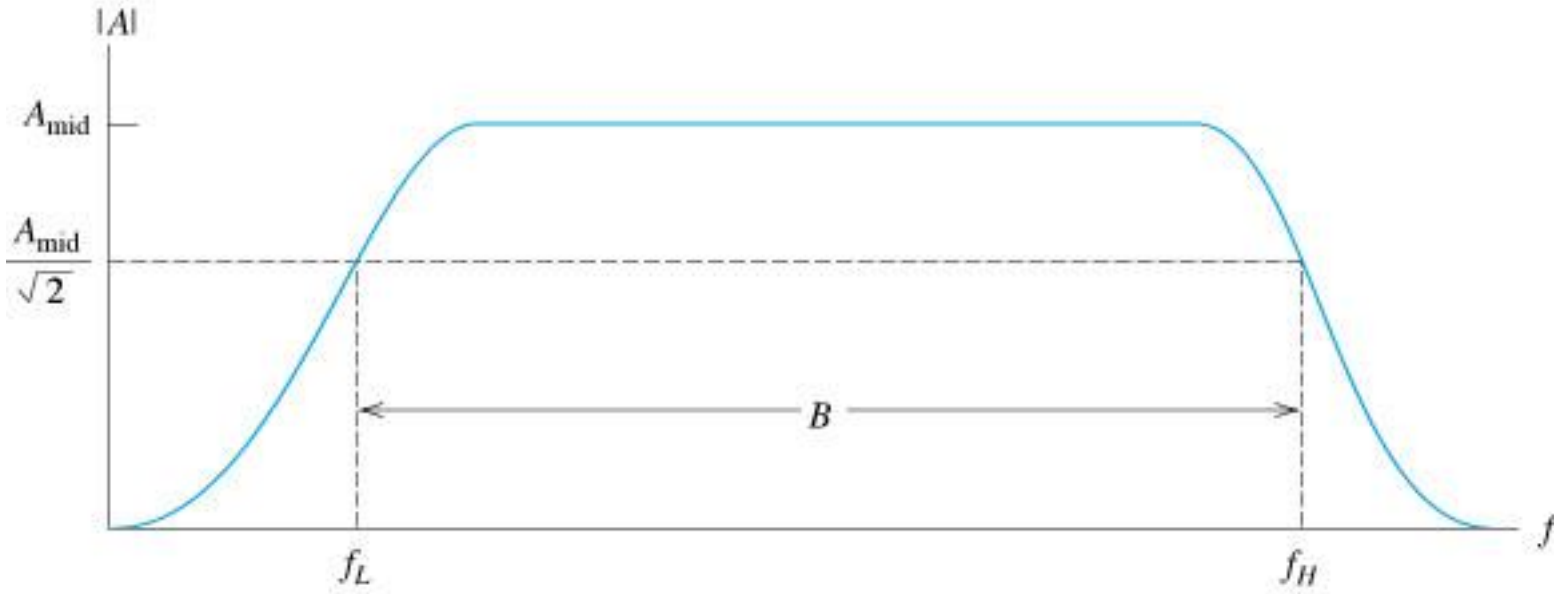


Figure 2.25 Bode plot of open-loop gain for a typical op amp.

Båndbredde

Med mindre andet er specificeret, menes der altid 3 dB frekvensen, når man snakker om båndbredde:



Figur 1.39

Gain versus frequency for a typical amplifier showing the upper and lower half-power (3-dB) frequencies (f_H and f_L) and the half-power bandwidth B.

Åbensløjfe forstærkningen A_{OL}

$$A_{OL}(f) = \frac{A_{0OL}}{1+j\frac{f}{f_{BOL}}}$$

Hvordan fås fasekarakteristikken for A_{OL}?

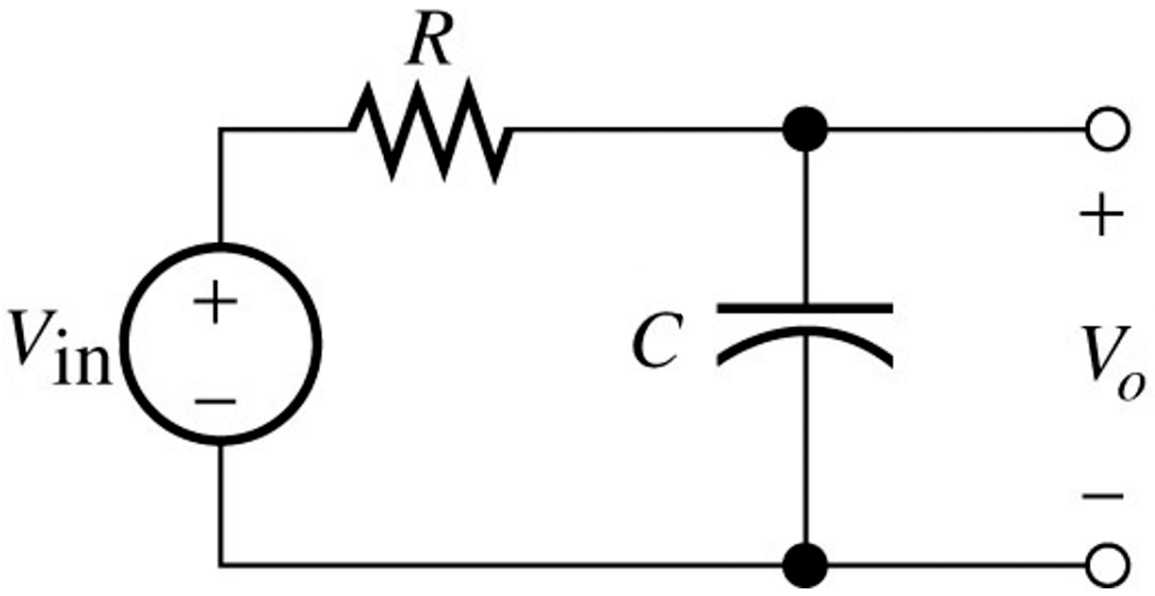
Åbensløjfe forstærkningen A_{OL}

$$A_{OL}(f) = \frac{A_{0OL}}{1+j\frac{f}{f_{BOL}}}$$

Hvordan fås fasekarakteristikken for A_{OL}?

Konklusion: $f >> f_{BOL} \Rightarrow -90^{\circ}$

Figur 8.1: Lavpas RC-filter

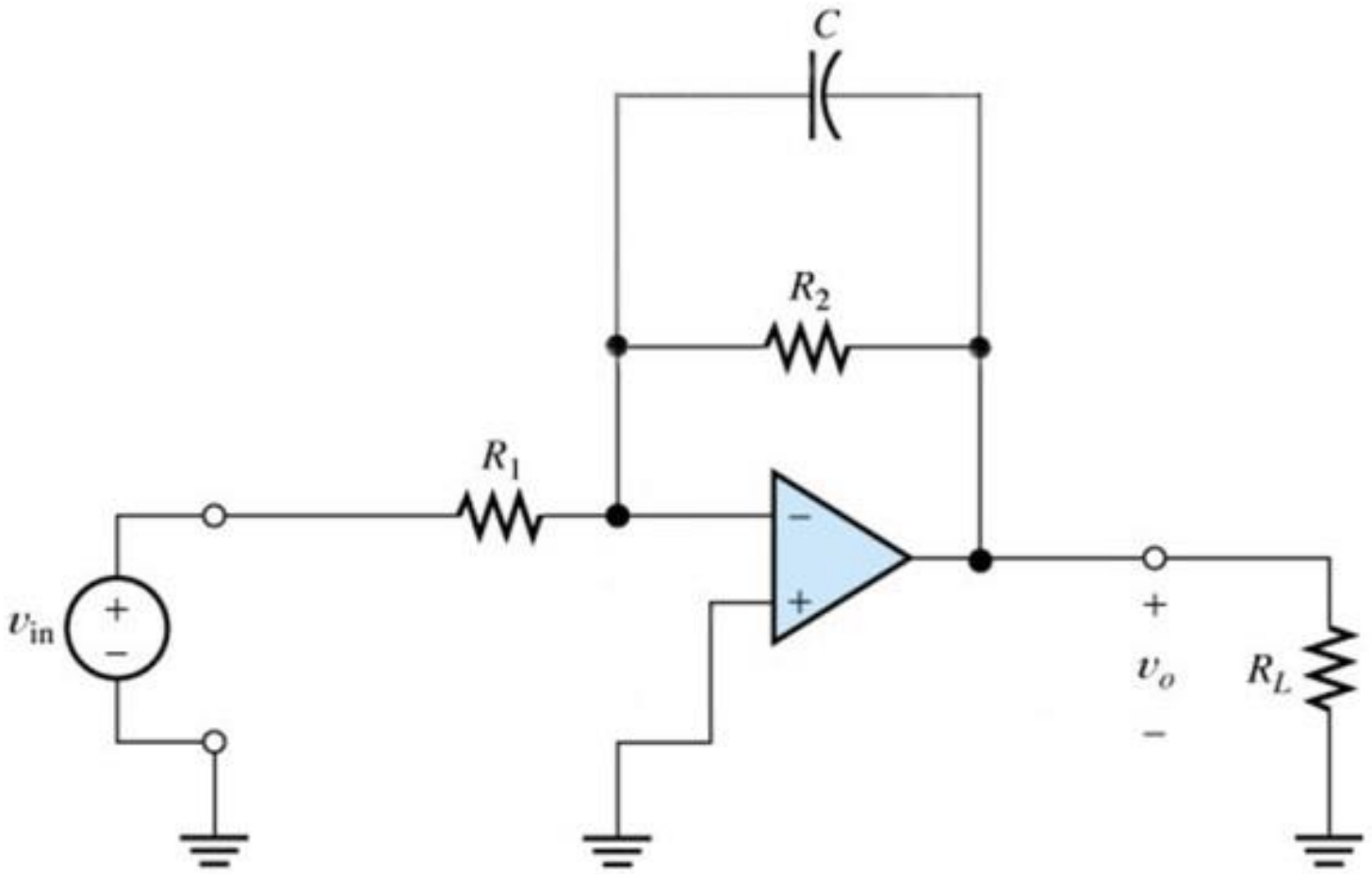


Overføringsfunktion?

Amplitudekarakteristik?

Sp. 3 fra forberedelsen:

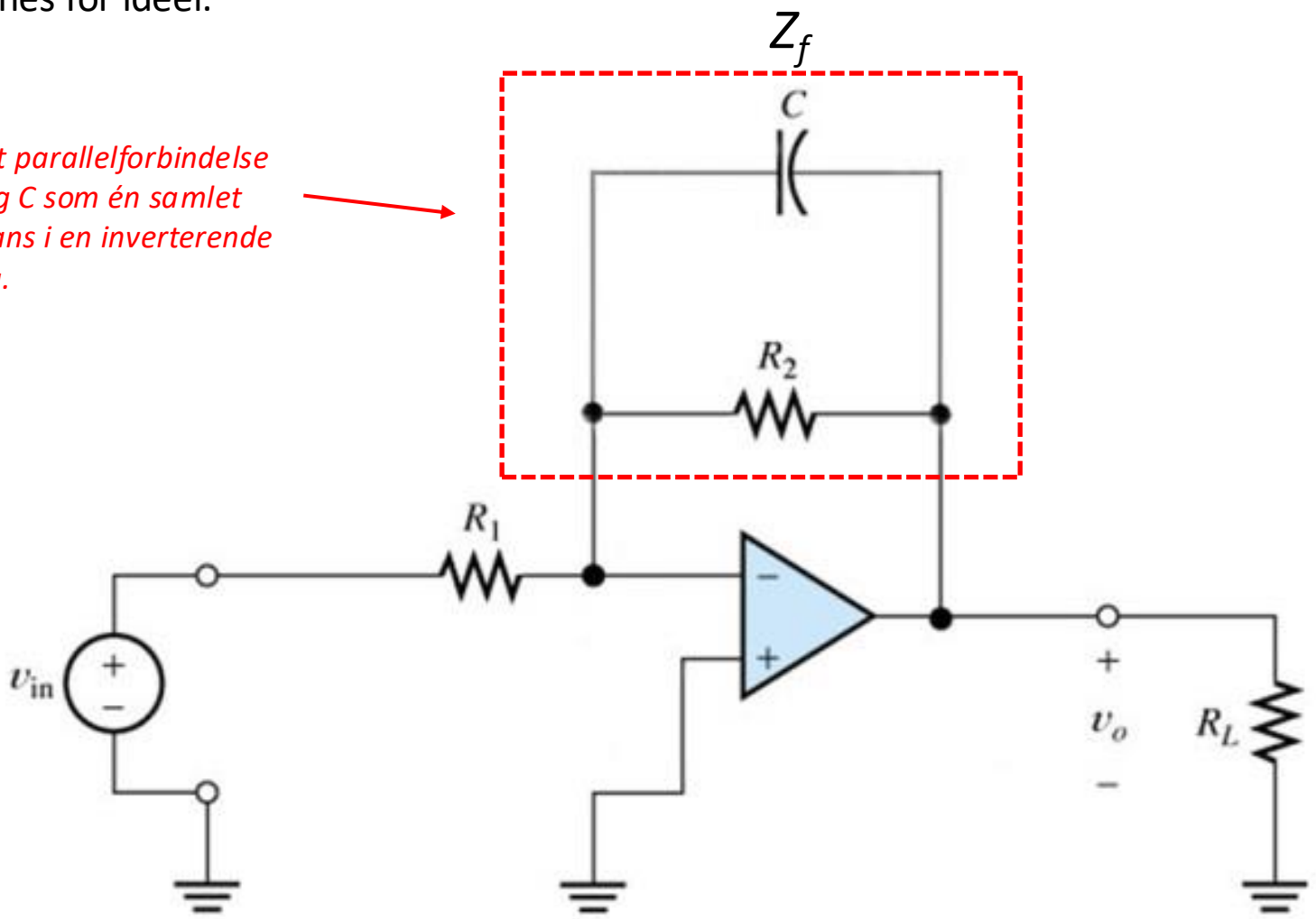
Betragt en inverterende forstærker som på figur 2.5. Placer en kondensator C parallelt med R_2. Find overføringsfunktionen $A_v = \frac{v_o}{v_{in}}$ og skitser Bode Plot for denne, når operationsforstærkeren kan regnes for ideel.



Sp. 3 fra forberedelsen:

Betragt en inverterende forstærker som på figur 2.5. Placer en kondensator C parallelt med R_2. Find overføringsfunktionen $A_v = \frac{v_o}{v_{in}}$ og skitser Bode Plot for denne, når operationsforstærkeren kan regnes for ideel.

Tip:
Betragt parallelforbindelse af R_2 og C som én samlet impedans i en inverterende kobling.



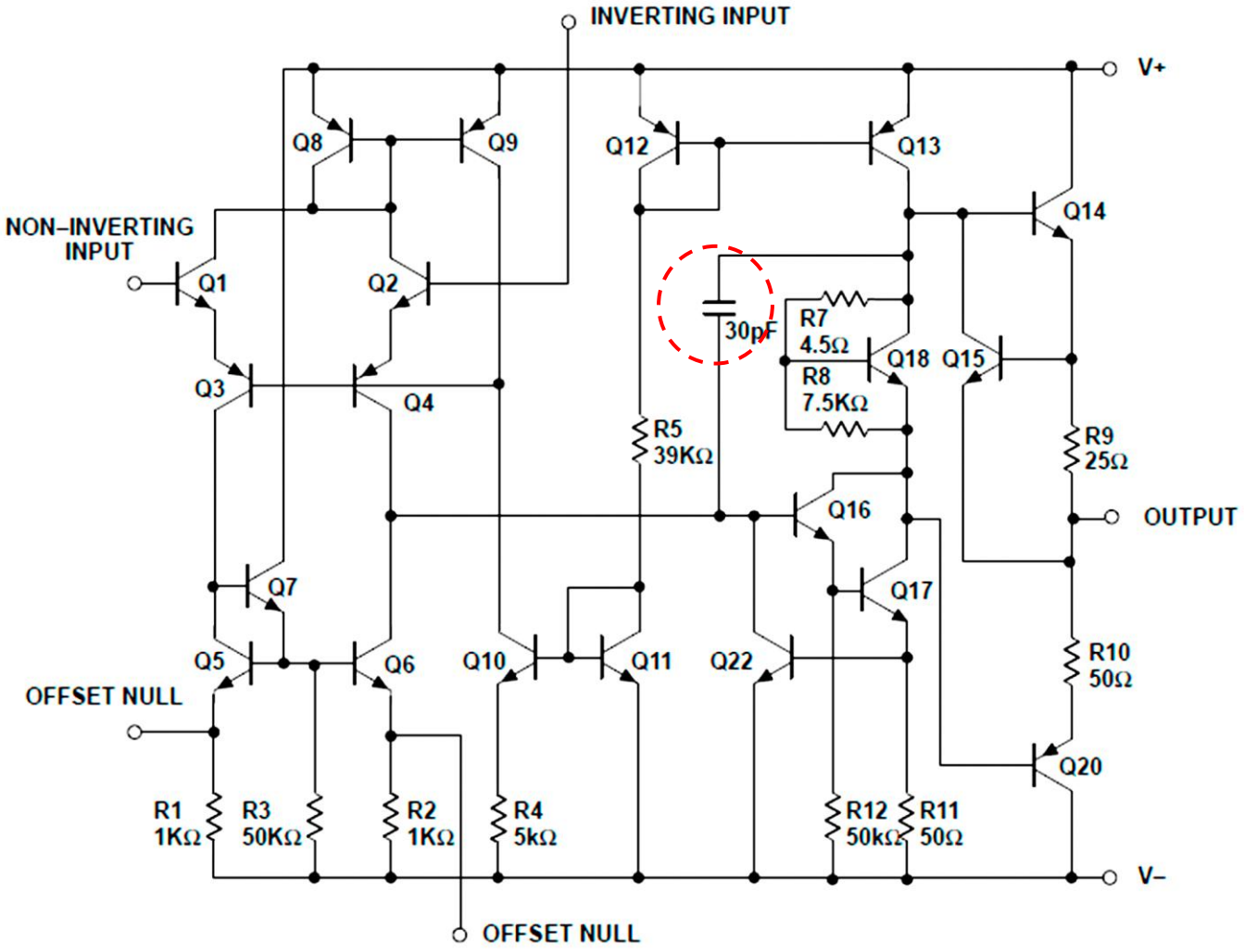


Diagram over µA741

C6

Pause bagefter!



*Frekvensforhold for **lukketsløjfeforstærkningen** A_{CL}*

Hvordan afspejler frekvensafhængigheden i A_{OL} sig i A_{CL}?

Hvordan finder man 3dB-båndbredden (f_{BCL}) for A_{CL}?

***K_f** frekvensafhængig:*

$$K_f = \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} \Rightarrow$$

***A_{CL}** frekvensafhængig:*

$$A_{CL} = \frac{\alpha}{\beta} \cdot K_f$$

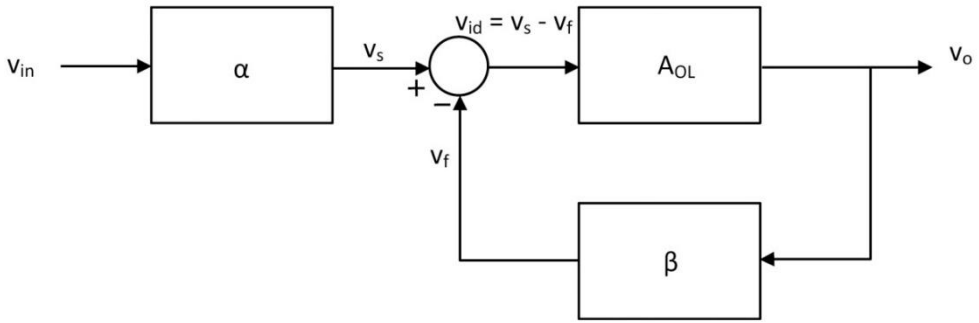
Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f



Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx ?$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx 1$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx 1$

$K_f \approx 1 \Rightarrow A_{CL} \approx ?$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx 1$

$K_f \approx 1 \Rightarrow A_{CL} \approx \frac{\alpha}{\beta} = A_{CL\ ideel}$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx 1$

3-dB frekvensen for A_{CL} findes altså der hvor:

$K_f = ?$

$K_f \approx 1 \Rightarrow A_{CL} \approx \frac{\alpha}{\beta} = A_{CL\ ideel}$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx 1$

$K_f \approx 1 \Rightarrow A_{CL} \approx \frac{\alpha}{\beta} = A_{CL\ ideel}$

3-dB frekvensen for A_{CL} findes altså der hvor:

$K_f = -3\ dB = \frac{1}{\sqrt{2}}$

Generel overføringsfunktion for modkoblet forstærker:

$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}} = \frac{\alpha}{\beta} \cdot K_f$$

Sløjfeforstærkningen $\beta \cdot A_{OL}$ er bestemmende for K_f

$|\beta \cdot A_{OL}| >> 1 \Rightarrow K_f \approx 1$

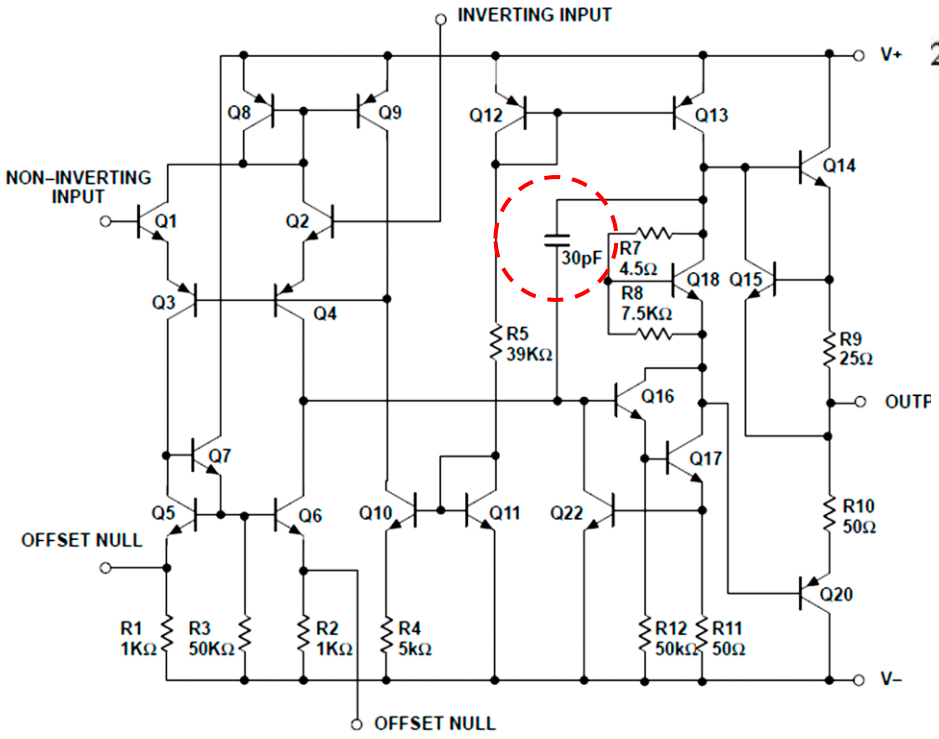
$K_f \approx 1 \Rightarrow A_{CL} \approx \frac{\alpha}{\beta} = A_{CL\ ideel}$

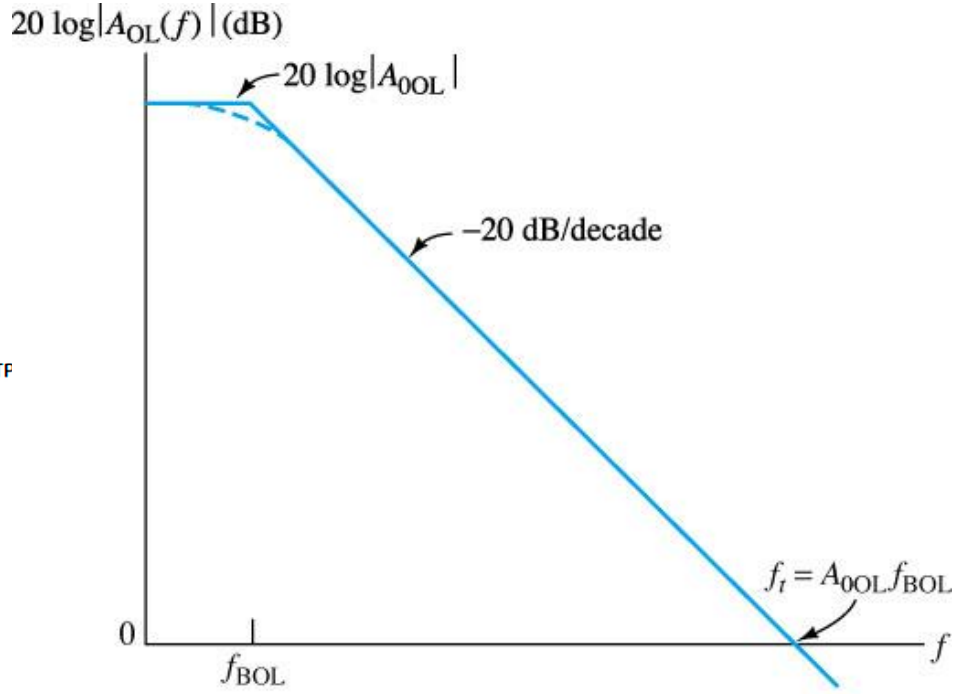
3-dB frekvensen for A_{CL} findes altså der hvor:

$K_f = -3\ dB = \frac{1}{\sqrt{2}}$

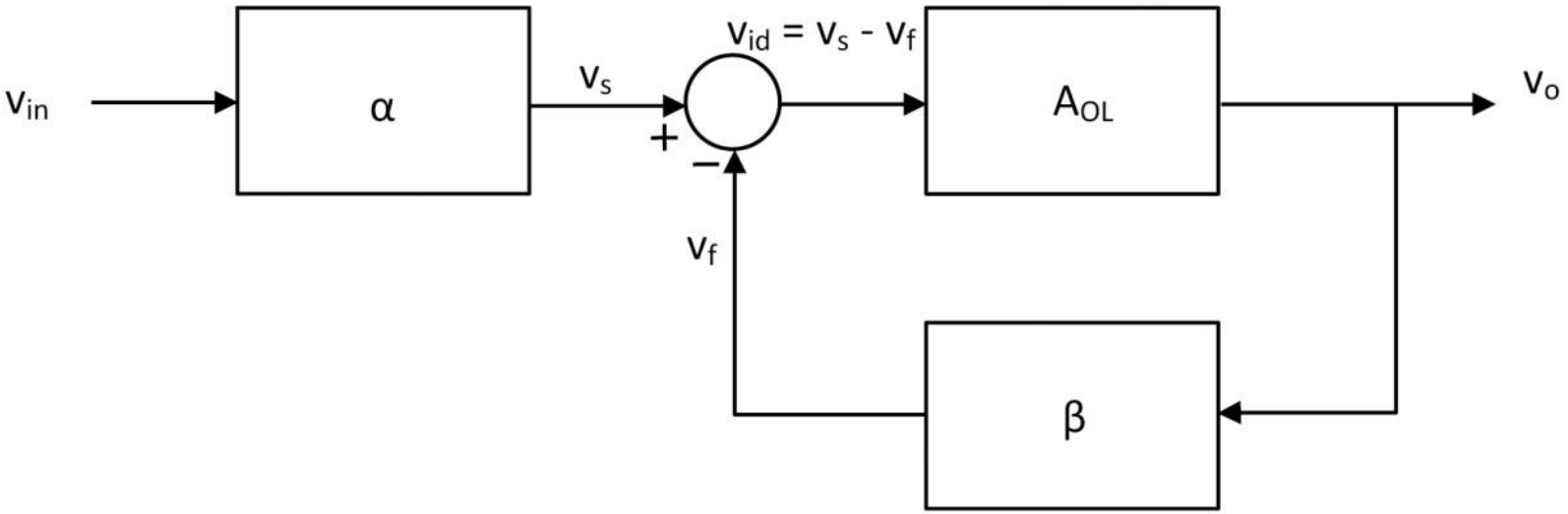
Grafisk overblik på et Bode plot - tavlen!

Hvorfor? 🤔


INVERTING INPUT
V+
NON–INVERTING INPUT
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
Q11
Q12
Q13
Q14
Q15
Q16
Q17
Q18
Q20
Q22
30pF
R7 4.5Ω
R8 7.5KΩ
R5 39KΩ
R9 25Ω
R10 50Ω
R1 1KΩ
R3 50KΩ
R2 1KΩ
R4 5kΩ
R12 50kΩ
R11 50Ω
OFFSET NULL
OFFSET NULL
V–

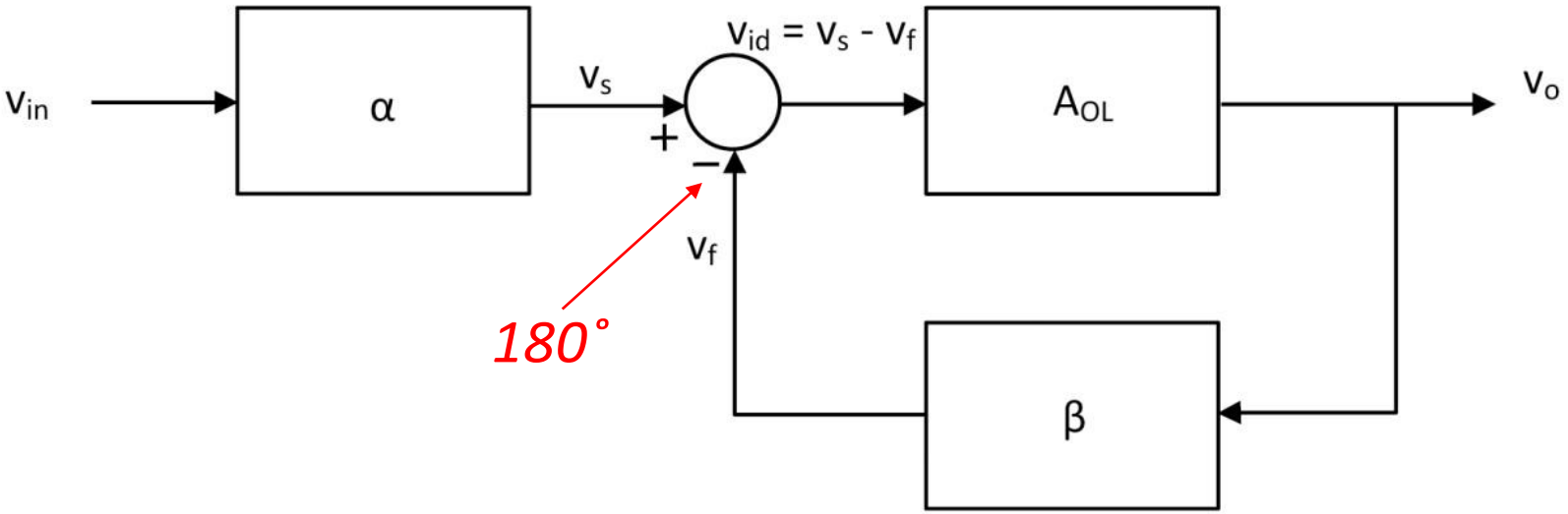

20 log|A_OL(f)| (dB)
20 log|A_0OL|
–20 dB/decade
f_t = A_0OL f_BOL
0
f_BOL
f

Modkoblet forstærker



Hvad er den tilsigtede (ideelle) fasedrejning rundt i feedbacksløjfen - altså fasedrejningen rundt i sløjfen $\beta \cdot A_{OL}$?

Modkoblet forstærker



Hvad er den tilsigtede (ideelle) fasedrejning rundt i feedbacksløjfen - altså fasedrejningen rundt i sløjfen $\beta \cdot A_{OL}$?

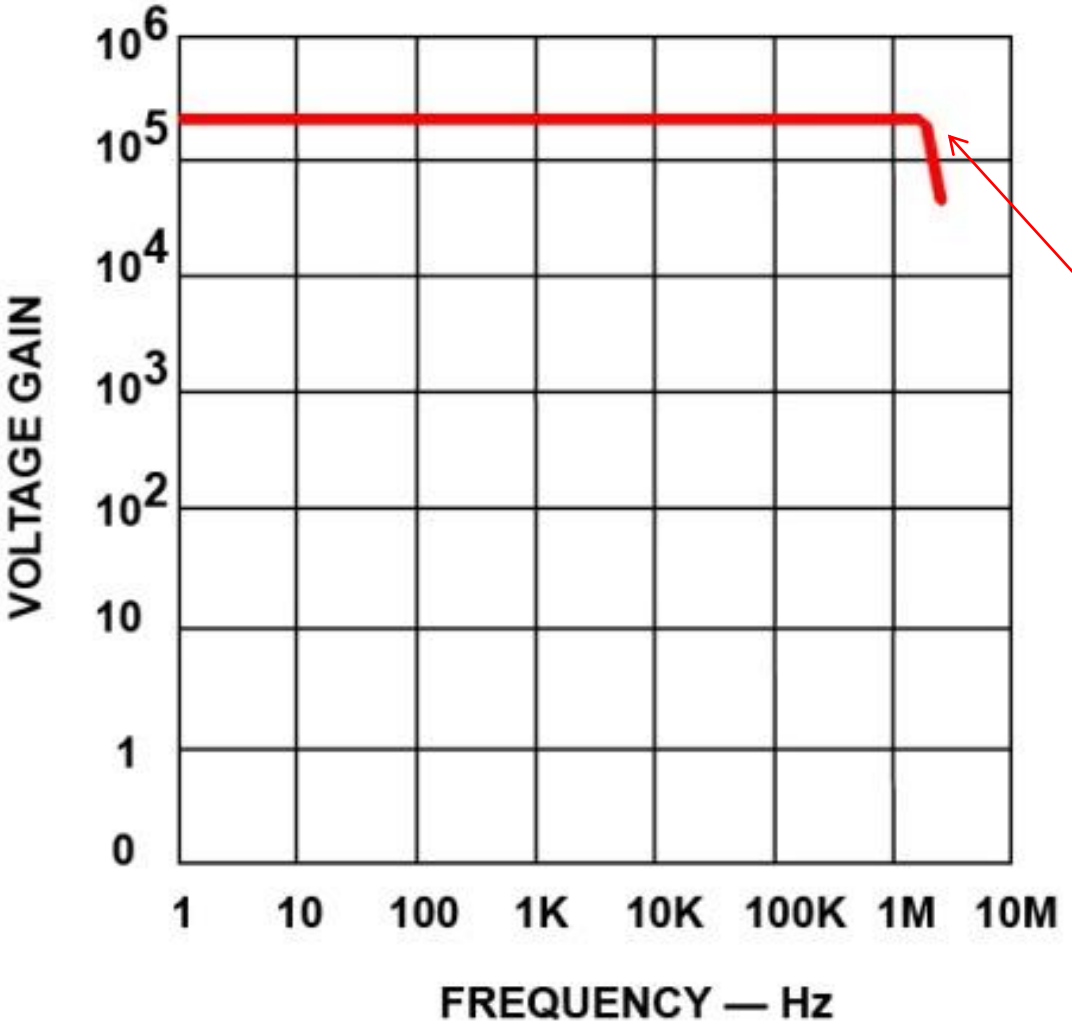
Vi husker lige…

$$A_{OL}(f) = \frac{A_{0OL}}{1+j\frac{f}{f_{BOL}}}$$

Konklusion: $f >> f_{BOL} \Rightarrow -90^\circ$

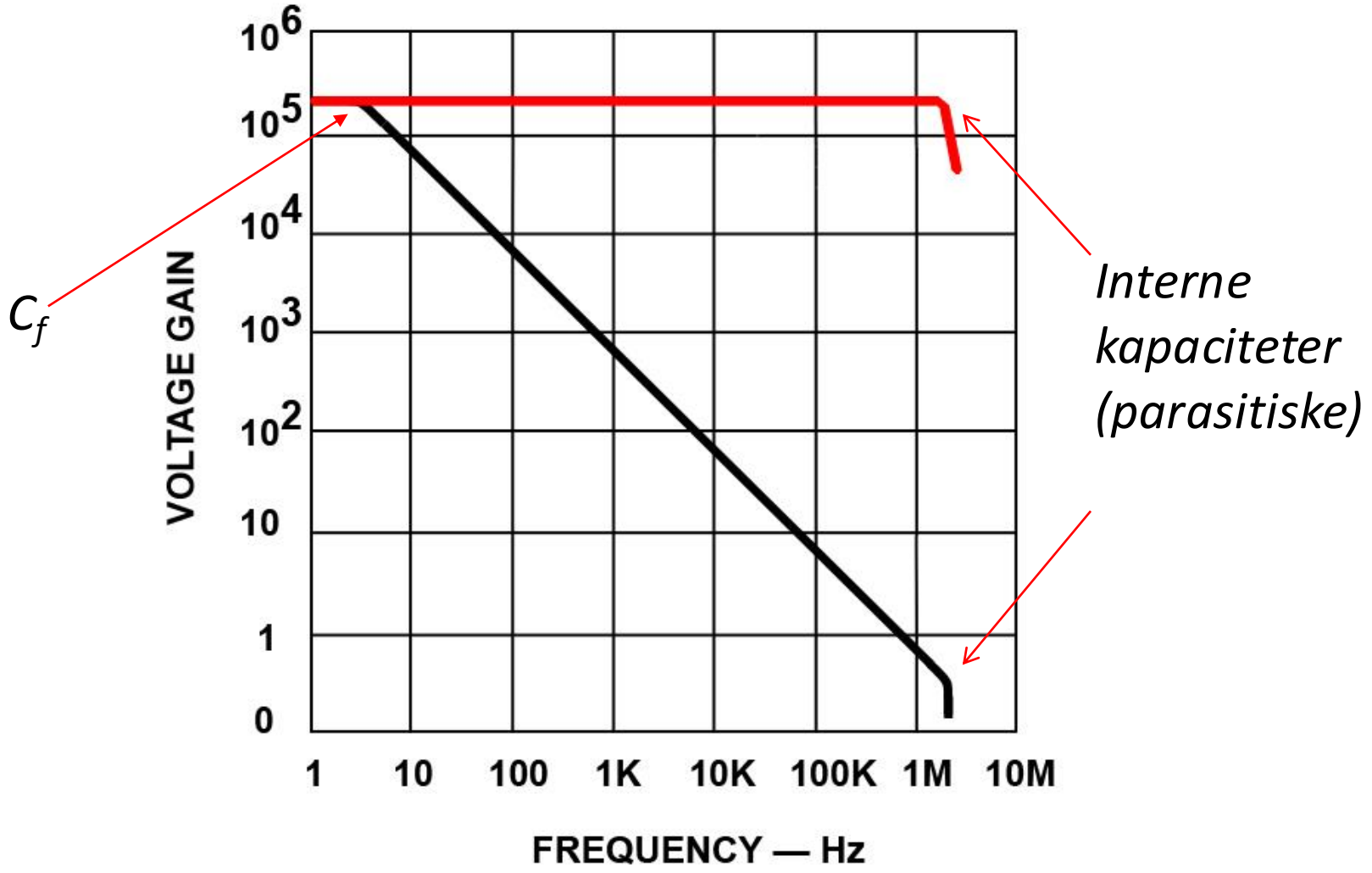
Hver pol/knækfrekvens giver -90° når man kommer længere op!

*Åbensløjfe forstærkning A_{OL} **uden** C_f*

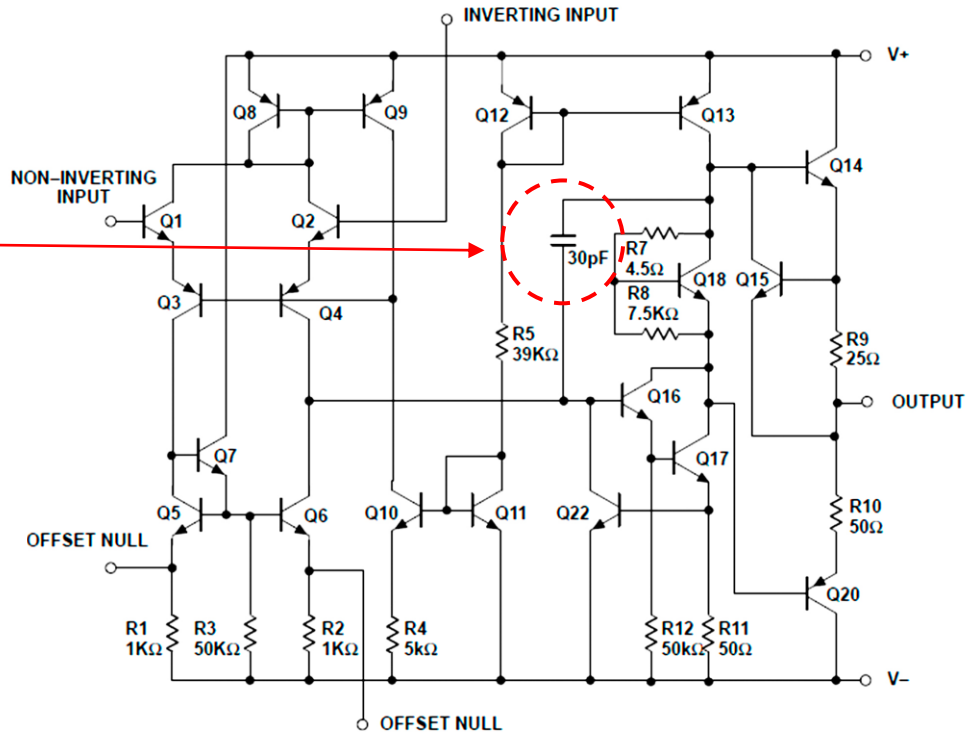


Interne kapaciteter (parasitiske)

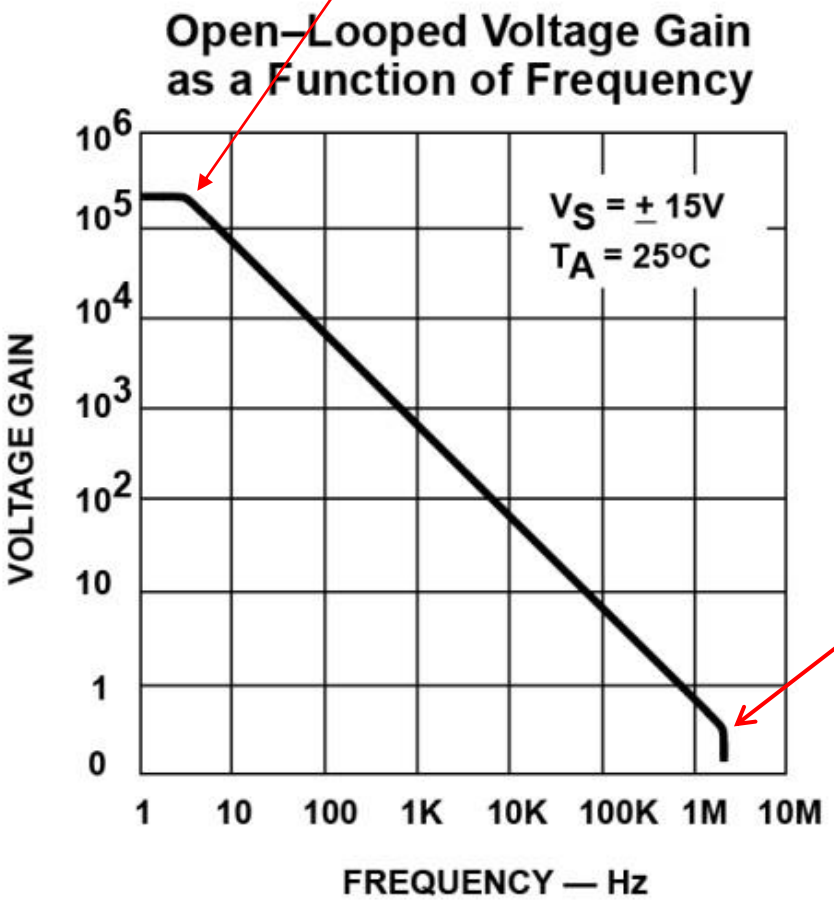
Åbensløjfe forstærkning A_{OL} med dominerende pol



Typisk A_{OL} for µA741



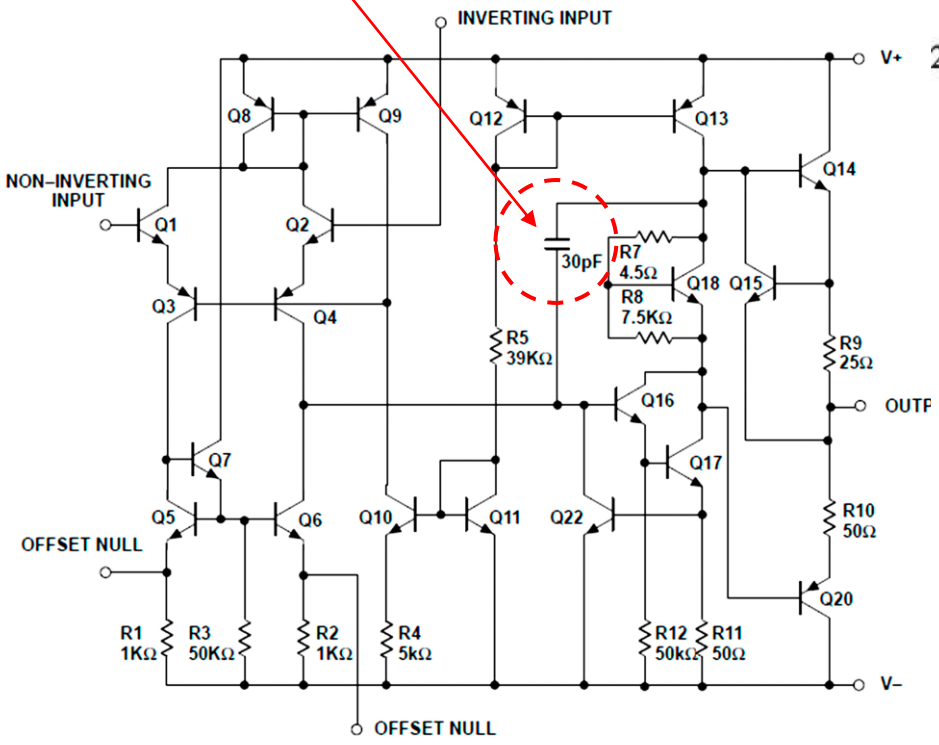
C_f

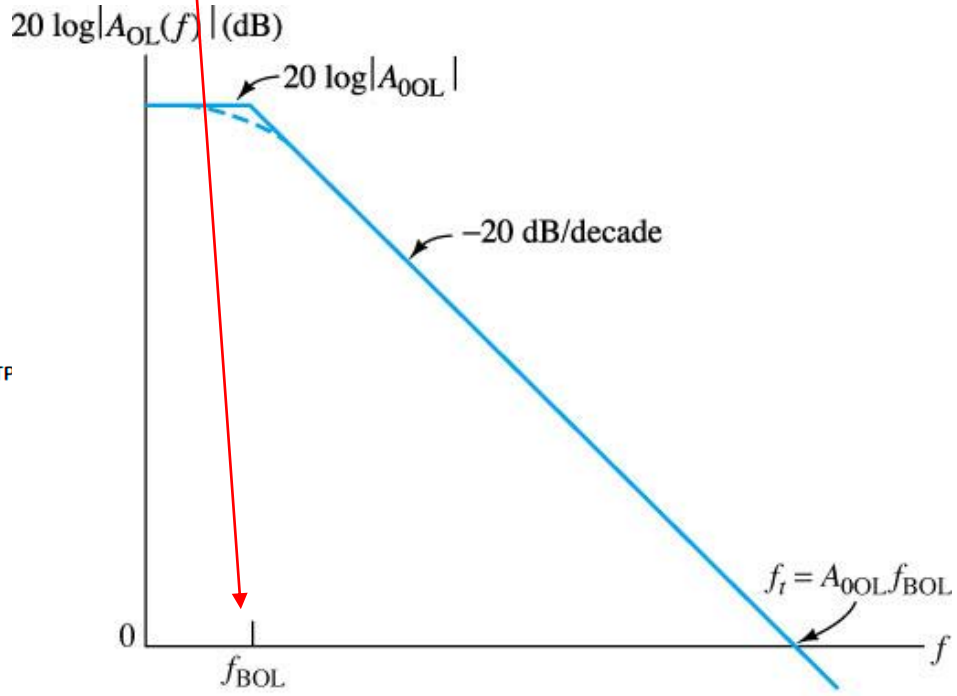


Interne kapaciteter (parasitiske)

Sp. 5 fra forberedelsen:

Nogle Op Amps er ***eksternt frekvenskompenserede***, dvs. man har placeret den frekvenskompenserende kondensator eksternt og har således mulighed for at variere f_{BOL}.
Hvilken fordel kan opnå ved dette?

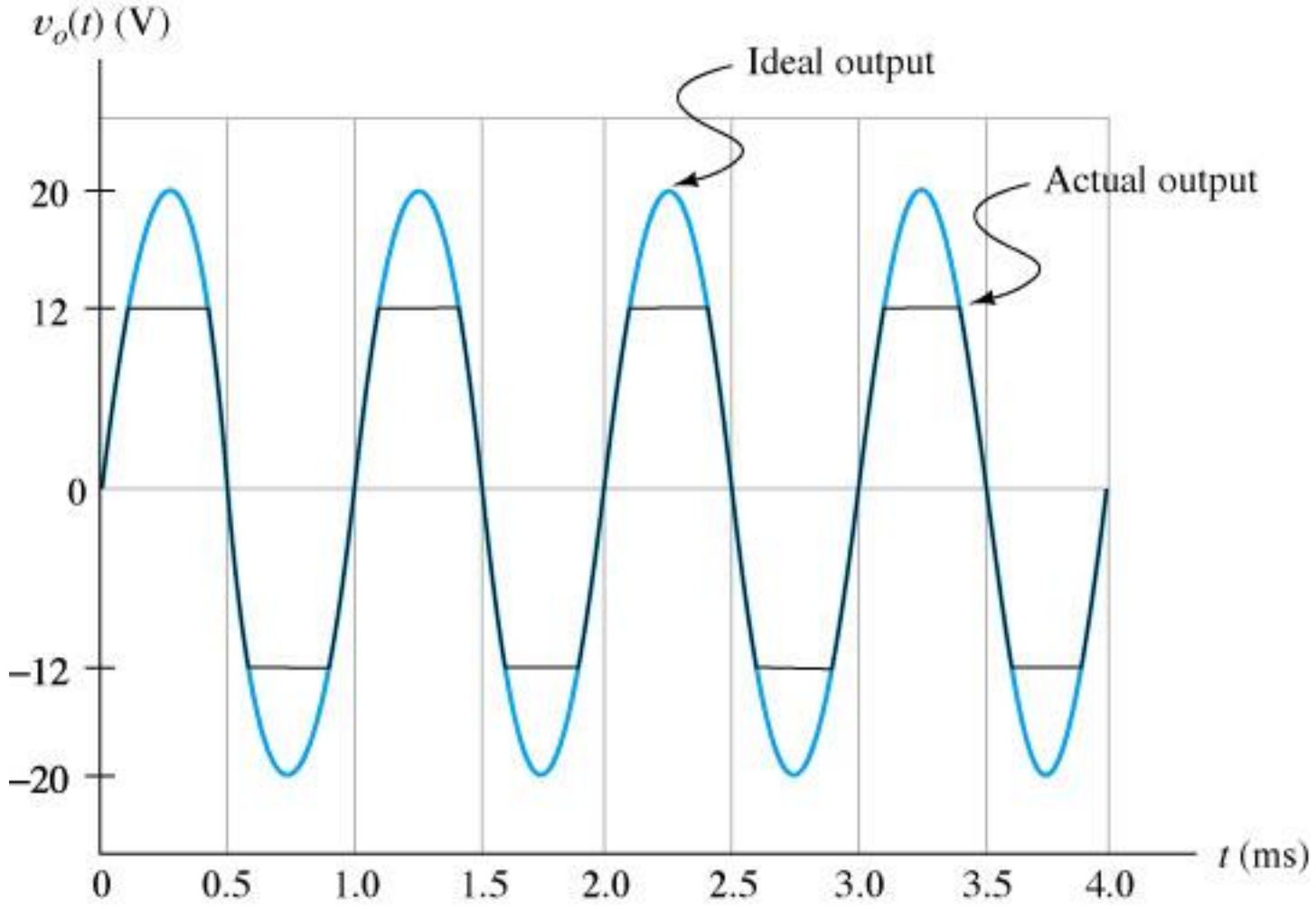




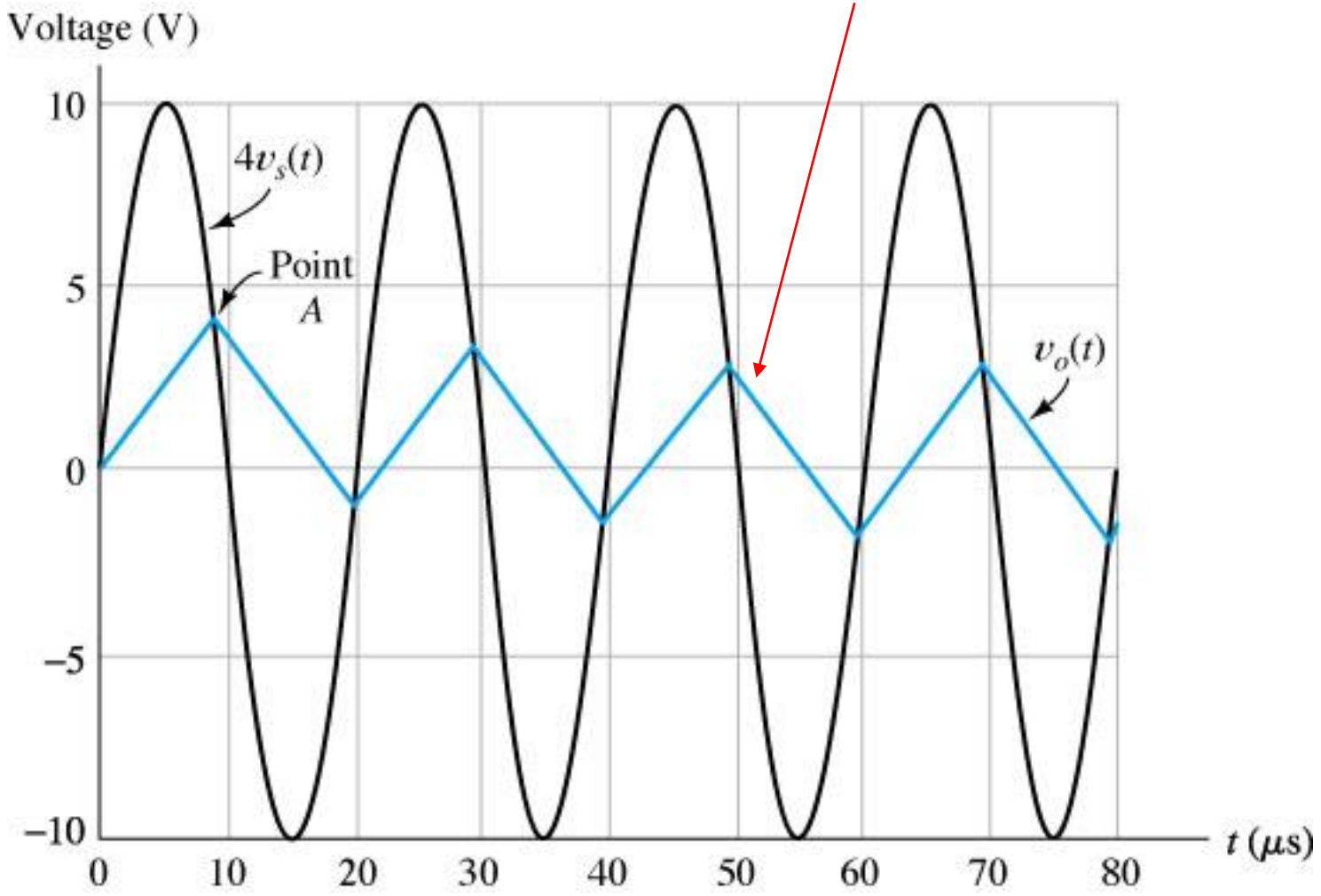
Hvorfor så dette?

Sp. 6 fra forberedelsen:

Hvad kan være årsagen til, at et udgangssignal "klipper"?

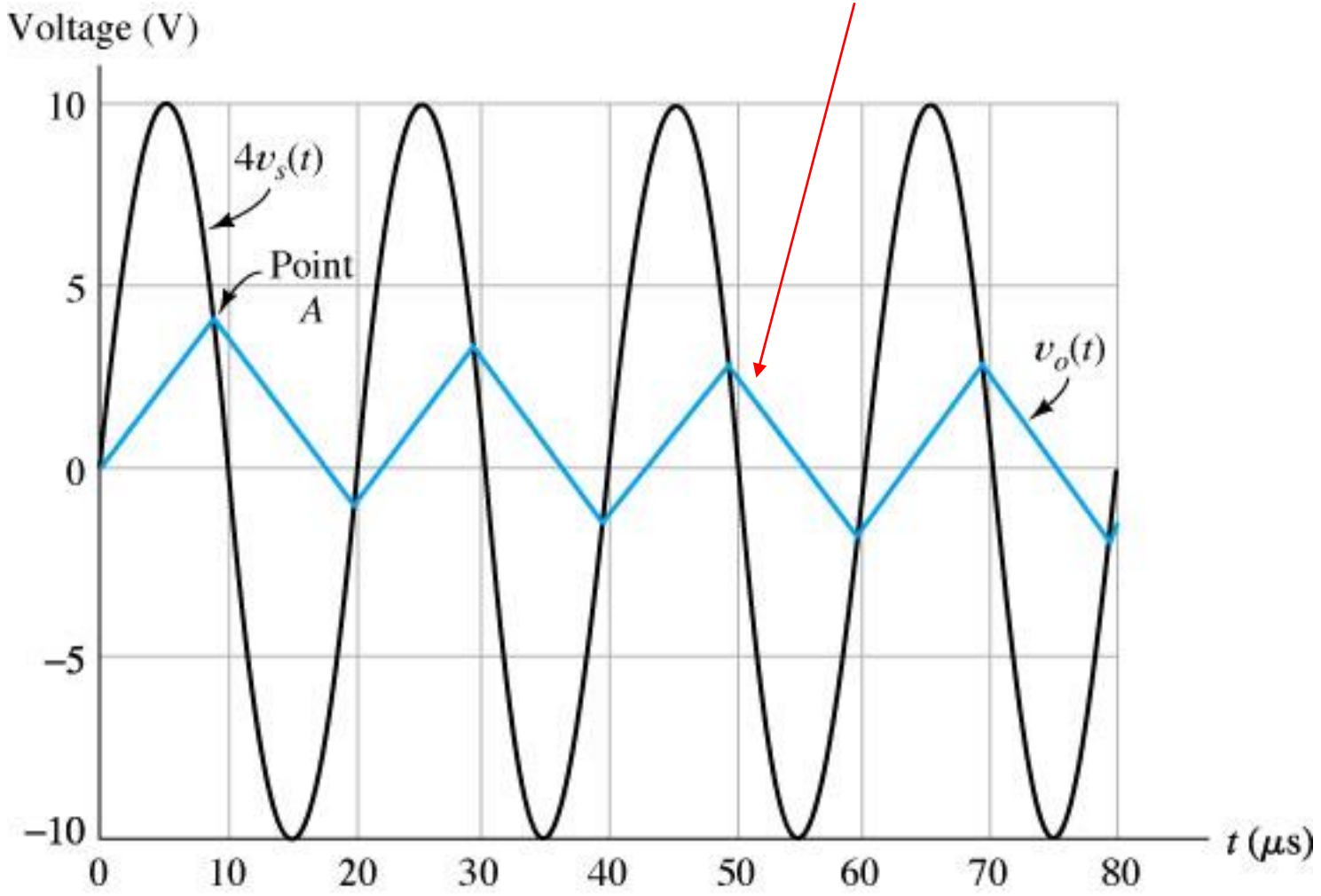


Hvad er så dette?



Figur 2.31

Hvad er så dette?



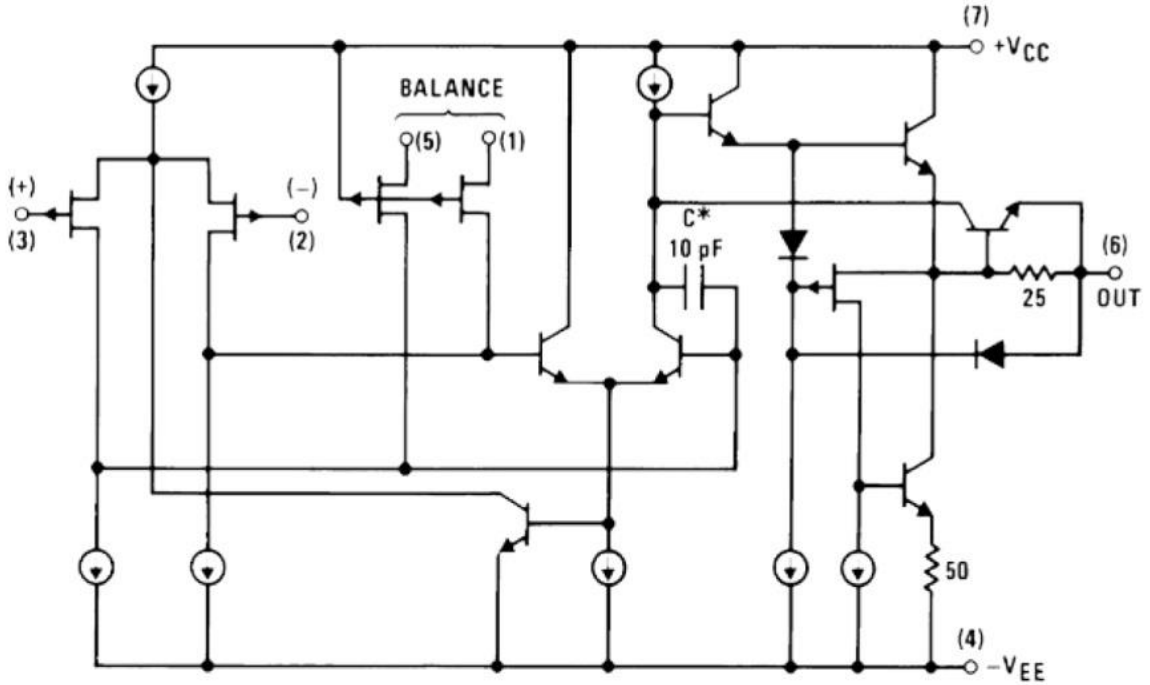
Figur 2.31

***Slewrate** begrænsning er også et **mætningsfænomén**!*

*Definition af **SR**?*

LF355/LF356/LF357

Simplified Schematic



*3pF in LF357 series.

BI-FET™, BI-FET II™ are trademarks of National Semiconductor Corporation.

Dynamiske parametre for LF355/LF356/LF357

AC Electrical Characteristics

$T_A = T_J = 25°C$, $V_S = \pm 15V$

Symbol	Parameter	Conditions	LF155/355	LF156/256/356B	LF156/256/356/LF356B	LF257/357	Units
			Typ	Min	Typ	Typ	
SR	Slew Rate	LF155/6: A_V=1, LF357: A_V=5	5	7.5	12	 50	V/µs V/µs
GBW	Gain Bandwidth Product		2.5		5	20	MHz
t_s	Settling Time to 0.01%	(Note 7)	4		1.5	1.5	µs
e_n	Equivalent Input Noise Voltage	R_S=100 f=100 Hz f=1000 Hz	 25 20		 15 12	 15 12	 $nV/\sqrt{Hz}$ $nV/\sqrt{Hz}$
i_n	Equivalent Input Current Noise	f=100 Hz f=1000 Hz	0.01 0.01		0.01 0.01	0.01 0.01	$pA/\sqrt{Hz}$ $pA/\sqrt{Hz}$
C_{IN}	Input Capacitance		3		3	3	pF

Dynamisk båndbredde contra Full-Power båndbredde

Dynamisk 3-dB båndbredde

(side 86)

Full-Power båndbredde

(side 93)

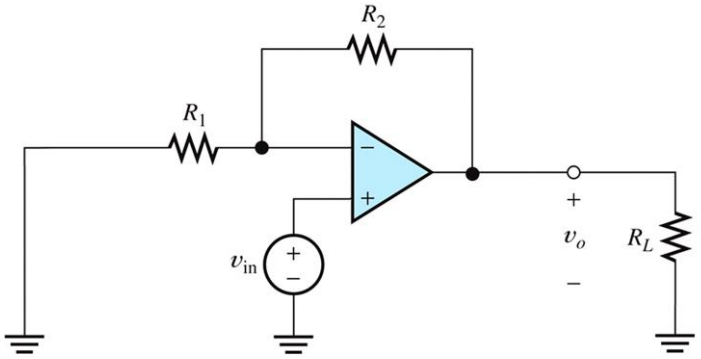
Forskel?

Dynamisk båndbredde contra Full-Power båndbredde

Dynamisk 3-dB båndbredde
(side 86)

Full-Power båndbredde
(side 93)

Forskel?



$v_{in} = 1 \cdot \sin(\omega t)$ [V]

$A_{0OL} = A_{OL}(0) = 120$ dB; $f_{BOL} = 10$ Hz

$SR = 5$ V/µs

$R_1 = 1$ kΩ

$R_2 = 9$ kΩ

$R_L = 47$ kΩ

Hvilken båndbredde bliver den begrænsende her?

Midtvejsdiskussion
tirsdag den 11. marts

C4